

Week 5 Practice Addendum

Teacher Guide, Pacing, and Full Answer Key

Student file: `Whately_Logic_Week5_Practice_Addendum.pdf`

Place: same folder as the Week 5 student reading

Why this packet exists

Week 5 packs mood, light figure, formal validity, and several named invalid patterns into one reading. The Logic Lab (**U1L5LL**) gives thin graded transfer practice. Most students need **volume** before transfer: many short labels, many moods, many valid/invalid calls.

This addendum is **formative practice**, not a second AI-graded lab. No Reading Focus. Full key lives here so you can teach from it.

What students should master (and what they should not)

Must own by end of Week 5

- Name A/E/I/O for ordinary sentences.
- On a three-line syllogism: find S, P, M; write mood major–minor–conclusion.
- Treat **Barbara**, **Celarent**, **Darii**, **Ferio** as the four strong **first-figure** home-base forms (names + moods + plain-English “why”).
- Spot invalid form, especially: undistributed middle; two negative premises; illicit process; negative premise with affirmative conclusion; particular evidence with universal conclusion.
- Run a **form test** without yet settling all facts (Week 6).

Do not require

- All 19 traditional valid moods, or the full mnemonic poem beyond the four first-figure names.
- Full reduction theory (consonants in medieval names).
- Figure numbers 2–4 as a memorized catalog (placement awareness is enough).
- Soundness / fact test as the main skill (that is Week 6).

Why Barbara, Celarent, Darii, Ferio (not “too much”)

These four are the classical **perfect** first-figure moods. They are enough to make “strong form” concrete without drowning students.

Name	Mood	Chain idea
Barbara	A-A-A	All S in M; all M in P $\Rightarrow$ all S in P
Celarent	E-A-E	All S in M; no M in P $\Rightarrow$ no S in P
Darii	A-I-I	Some S in M; all M in P $\Rightarrow$ some S in P
Ferio	E-I-O	Some S in M; no M in P $\Rightarrow$ some S not P

First-figure layout: major premise has M as subject (M–P); minor has M as predicate (S–M). That is the clean chain. Teaching the names is worth it because students finally have four “this is what good looks like” models. Spelling: **Celarent**, **Darii**, **Ferio** (not Celerent / Dario / Fario).

Suggested pacing

- **Day 1:** Student reading + Reading Focus discussion (not this packet).
- **Day 2:** Addendum Parts A–C in class (pairs OK). Spot-check moods on the board.
- **Day 3:** Part D (four strong forms) whole-class; start Part E.
- **Day 4:** Finish E–G; homework H or I if needed.
- **Day 5:** Part I debrief; Part J memory card; then assign graded **U1L5LL**.
- **Later:** Lit Example U1L5LE.

If you only have two practice days: do A, B (half), D, E (half), I. Skip F–G or use as challenge.

Grading options

- **Recommended:** completion + accuracy spot-check (e.g. grade Part E odds and Part I).
- Light points: Part B (12), Part D (name match), Part E (12), Part I (4) — teacher choice.
- Do **not** put this packet into the AI answer-key pipeline unless you later want a formal key file; U1L5LL remains the graded transfer product.

Common confusions (teach these early)

1. **Mood order:** always major premise, minor premise, conclusion — not “order on the page” if the paragraph scrambles them (see C6, Part I).
2. **True conclusion $\neq$ valid form.** Preview Week 6; kill this habit now.
3. **Every invalid argument is not “undistributed middle.”** Force the flaw bank.
4. **O vs E:** “Some S are not P” is O, not E.
5. **Singulars:** “Maya is ...” treated as A-style for mood (standard intro-logic classroom move).
6. **Illicit process vs undistributed middle:** middle fails to link vs a term distributed in the conclusion but not in its premise.
7. **Two negatives:** two separations from a third class do not relate the first two classes.

Relation to other Week 5 products

- Reading: concepts + few models. Addendum: volume. Lab: sparse graded transfer. LE: literary application.
- Examples here avoid the reading’s whales / tyrannical orders / careful proofs / Brutus-Macbeth form demos and avoid U1L5LL’s Route 9 / maples / podcasts / senators / Priya set (Part E uses roses/tulips, streaming/assemblies, federal judges instead).

Sources / pedigree (for you, not for students)

Classic first-figure illustrations modernized from standard categorical-syllogism teaching (Aristotelian first figure; medieval names Barbara, Celarent, Darii, Ferio; common invalid patterns: undistributed middle, exclusive premises, illicit major/minor). School/civic wording is original classroom adaptation.

Full answer key

Part A — Proposition warm-up

1. A
2. E
3. I
4. O
5. A (“Every” = All)
6. E (“None ... is” = No)
7. I (“At least one” = Some)
8. O (“Not every volunteer finished” $\approx$ Some volunteers are not people who finished the training module)

Part B — Full labels + mood

Accept reasonable synonymy in term wording (e.g. “dogs” vs “all dogs”).

- B1** S: dogs P: warm-blooded animals M: mammals Mood: **A-A-A** (Barbara shape)
- B2** S: snakes P: fur-bearing animals M: reptiles Mood: **E-A-E** (Celarent)
- B3** S: night-shift workers P: citizens with a current address on file M: registered voters Mood: **A-I-I** (Darií)
- B4** S: stickers on that windshield P: valid dashboard displays M: expired parking permits Mood: **E-I-O** (Ferio)
- B5** S: fire alarms in this building P: procedures that clear the stairwells M: emergency drills Mood: **A-A-A**
- B6** S: Jordan’s lunch holds P: cleared lunch account(s) M: unpaid cafeteria debt(s) Mood: **E-A-E**
- B7** S: pool staff members P: people trained in CPR M: certified lifeguards Mood: **A-I-I**
- B8** S: rooms on the first floor P: open student workspaces M: locked server closets Mood: **E-I-O**
- B9** S: Maya P: student with a complete application M: scholarship finalist(s) Mood: **A-A-A** (singulars as A)
- B10** S: this powder P: allowed competition aid M: banned substance Mood: **E-A-E** (conclusion negative universal/singular)
- B11** S: the tablet on Row C P: (things that) must be returned before last period M: tablet(s) checked out from the cart Mood: **A-A-A**
- B12** S: Route 14 P: (routes that) stop at the north lot M: late bus route(s) Mood: **E-A-E**

Part C — Mood only

1. A-A-A
2. E-A-E
3. A-I-I
4. E-I-O

5. A-E-E (major A, minor E, conclusion E — valid in some figures; do not demand figure name; mood letters only here)
6. **Careful:** Conclusion is about English assignments / papers on time. Premises as printed: O then A then O. If students reorder to standard major-first: major may be treated as the premise with “papers turned in on time” as P. Accept **O-A-O** if they keep print order and note scramble; best teaching move: rewrite as Major: All of these essays are English assignments. (A)
Minor: Some essays are not papers turned in on time. (O)
Conclusion: Some English assignments are not papers turned in on time. (O)
That is messy distribution-wise; use this item to teach **find conclusion first**, not as a clean validity case. For mood-as-printed: **O-A-O**.
7. A-E-E
8. O-A-O (print order: O major-looking, A, O) — actually: “Some campus maps are not up-to-date guides” (O), “All campus maps are free handouts” (A), conclusion O. Mood as printed **O-A-O**.
9. A-A-A
10. E-I-O

Teacher tip: If C5–C8 slow the class, skip them and return after Part D.

Part D — Four strong forms

D1

1. Barbara A-A-A
2. Celarent E-A-E
3. Darii A-I-I
4. Ferio E-I-O

D2

1. Conclusion: All yearbook editors are people who verify quotations.
2. Conclusion: No bike in Rack 3 is a bike left safely overnight.
(or: None of the bikes in Rack 3 is left safely overnight.)
3. Minor: Some seniors are certified tutors.
(Any I-premise that places some seniors inside certified tutors works.)
4. Major: No peanut dessert is a safe snack for that allergy table.
(E premise excluding peanut desserts from the safe-snack class.)

D3 — sample strong answers

1. First figure lines S up under M and M under P (or outside P in Celarent). If premises are granted, the conclusion only reports the link already forced — not a new leap.
2. Darii/Ferio still **force** a particular conclusion from a universal major + particular minor. “Some” is not a guess; it is the largest conclusion those premises can guarantee. Overclaiming “all” would be the weak move.

Part E — Valid or invalid

1. Mood A-A-A. **Invalid.** Undistributed middle (“plants” never links tulips into roses).
2. Mood E-E-E. **Invalid.** Two negative premises.
3. Mood A-E-E. **Invalid.** Illicit process of the major term (“government officials” distributed in conclusion, not properly distributed in major A premise).
4. Mood A-A-A. **Valid.** First-figure Barbara shape.
5. Mood E-A-E. **Valid.** Celarent shape.
6. Mood A-I-I but middle is “students with community hours” as predicate of both — **Invalid.** Undistributed middle (both premises place groups inside community-hours students; does not force juniors into honor society).
Note: If written as major “All honor society. . .”, minor “Some juniors are students with community hours,” the middle is not even shared as a linking middle in first-figure form — still invalid.
7. Mood E-I-O. **Valid.** Ferio shape (first figure).
8. Mood A-E-E. Often taught as valid **Camestres**-like second figure if middle placement is P-side; many intro courses accept **valid**. If students say invalid from first-figure-only habits, discuss placement: major A (scored above 90 = M subject), minor E (invited = M subject) — actually both have “invited to review dinner” linking differently. Walk carefully:
 Standard form attempt: All above-90 are invited. No invited is barred. Therefore no above-90 is barred. This is valid (chain of exclusion). Mark **Valid**.
9. Mood I-A-A as printed. **Invalid.** Particular premise cannot yield universal conclusion about all bus riders (over-strong conclusion). Also order is scrambled.
10. Mood E-E-E. **Invalid.** Two negative premises.
11. Mood A-A-A. **Invalid.** Undistributed middle (“ceiling fixtures”).
12. Mood A-I-I. **Valid.** Darii shape.

Part F — Same shape, different subject**F1**

Both are A-A-A inclusion chains (first-figure style if ordered M–P / S–M). **Valid** form. Shows same shape can carry different topics.

F2

Undistributed middle (both groups under a larger class). **Invalid.** Same bad shape, different nouns.

F3

Two negative premises. **Invalid.** Same failure mode in biology wording and school wording.

Part G — Light figure

1. M subject of major, M predicate of minor. **First figure: Yes.**
2. Middle: people who stretch before practice (or “stretch before practice” class). M is **predicate** of both premises. **First figure: No.** (Classic undistributed-middle shape.)

3. Middle: reptiles. M subject major, M predicate minor. **First figure: Yes.** (Celarent)
4. Middle: smartphones. M subject major; M predicate minor? Minor is “No wall clocks are smartphones” — M is predicate. First figure pattern for E minor is different; still M-predicate in minor. Major has M as subject. **First figure-ish layout** but mood A-E-E with this placement is **invalid** (illicit major: “devices that use batteries” over-extended). Mark **Invalid**; illicit process of major.

Part H — Repair (accept equivalent fixes)

1. Classic undistributed middle. Repair examples: change conclusion to something not forced — or change a premise to a real first-figure link (e.g. All surgeons are nurses — false but valid Barbara if paired with All nurses work at hospital — still may not match intent). Best student move: **explain no safe patch to “all surgeons are nurses” from shared workplace alone**; or weaken to “Some people who work at the hospital are surgeons; some are nurses” (not a syllogism repair). Full credit for clear “shared large class does not identify subclasses.”
2. Two negatives about captains do not relate sophomores to freshmen. Explain cannot prove that conclusion; any “repair” must change the question.
3. Illicit major: A major does not distribute “students with strong essays.” Repair: weaken conclusion to something not claiming all transfer students lack strong essays — e.g. conclude only that no transfer student is a **scholarship winner** (but that needs different premises). Best: show original conclusion overclaims.
4. Particular + universal cannot prove all club members barred. Repair conclusion to: **Some club members are people barred from the party** (if chain is fixed in first figure: All who skipped are barred; some club members skipped $\Rightarrow$ some club members barred).
5. Affirming the consequent / wrong direction (undistributed middle on “events that empty the building”). Repair: cannot conclude fire drill from emptying alone; need “All events that empty the building are fire drills” (false) or drop the conclusion.

Part I — Buried syllogisms

I1

Major: Every athlete who failed the concussion screen sits out the next game. (A)

Minor: Rivera failed the concussion screen. (A singular)

Conclusion: Rivera sits out the next game. (A singular)

Mood: **A-A-A. Valid** (Barbara with singulars). Ignore snacks/popcorn distractors.

I2

Major: Nobody without a medical pass may use the elevator before 3:00. $\approx$ No person without a medical pass is a person who may use the elevator before 3:00. (E)

Minor: Sam has no medical pass. (A singular about Sam)

Conclusion: Sam may not use the elevator before 3:00. (E/A singular negative)

Mood: treat as **E-A-E. Valid** Celarent-style. Paint smell is distractor.

I3

Major: All students who complete six college essays are applicants ready for early decision. (A)

Minor: Jordan completed six college essays. (A singular)

Conclusion: Jordan is an applicant ready for early decision. (A singular)

Mood: **A-A-A. Valid in form** (facts/definitions may be doubtful — Week 6 point if students raise it; form is fine).

I4

Both premises put groups inside “pay attention.” Conclusion identifies the two groups. Mood A-A-A with undistributed middle. **Invalid.** Undistributed middle.

Part J — Quick reference (targets)

1. Mood = A/E/I/O pattern of major, minor, conclusion in that order.
2. First figure: major M–P; minor S–M; conclusion S–P.
3. Barbara A-A-A; Celarent E-A-E; Darii A-I-I; Ferio E-I-O.
4. Middle term never covers the whole class needed to link S and P.
5. Two exclusions from a third class do not relate the first two classes.
6. Form test: Does the conclusion follow from the premises?
7. Fact test: Are the premises true/justified? (Week 6)

Board scripts (3 minutes each)**Undistributed middle**

Draw one big circle (“people who stretch”). Two smaller circles inside that do not necessarily overlap (“varsity runners,” “managers”). Say: both inside big circle $\neq$ same small circle.

Two negatives

“No freshmen are captains. No sophomores are captains.” Both groups outside captains. They might still overlap each other or not. Nothing follows about freshmen vs sophomores.

Barbara chain

Three nested circles or a chain on the board: $S \subset M \subset P \Rightarrow S \subset P$.

Celarent chain

$S \subset M$, and M entirely outside $P \Rightarrow S$ outside P .

If students still drown

- Temporary rule: only first-figure examples until Part E is stable.
- Drop figure numbers entirely; keep only “chain vs shared-basket.”
- Use Part D names as the only “valid models” for a week, then expand.
- Do not split the unit number; extend calendar days inside Week 5 instead.

End of Teacher Guide — Week 5 Practice Addendum